of an edge of the cube then intersect along a circle, as they should, and other pairs of tori intersect only at the 0-cell of Z.

This model of Z has the advantage of displaying the symmetry group of the cube, a group of order 48, as a symmetry group of Z, corresponding to the symmetries of X permuting the three $S^1 \vee S^1$ factors and the two S^1's of each $S^1 \vee S^1$. Undoubtedly Z would be very pretty to look at if we lived in a space with enough dimensions to see all of it at one glance.

It might be interesting to see an explicit set of maps $S^2 \to Z$ generating $\pi_2(Z)$ as a $\mathbb{Z}[\pi_1]$-module. One might also ask whether there are simpler examples of these nonfinite generation phenomena.

Exercises

1. Show directly that if X is a topological group with identity element x_0, then any two maps $f, g : (Z, z_0) \to (X, x_0)$ which are homotopic are homotopic through basepoint-preserving maps.

2. Show that under the map $\langle X, Y\rangle \to \mathrm{Hom}(\pi_n(X, x_0), \pi_n(Y, y_0))$, $[f] \mapsto f_*$, the action of $\pi_1(Y, y_0)$ on $\langle X, Y\rangle$ corresponds to composing with the action on $\pi_n(Y, y_0)$, that is, $(\gamma f)_* = \beta_\gamma f_*$. Deduce a bijection of $[X, K(\pi,1)]$ with the set of orbits of $\mathrm{Hom}(\pi_1(X), \pi)$ under composition with inner automorphisms of π. In particular, if π is abelian then $[X, K(\pi,1)] = \langle X, K(\pi,1)\rangle = \mathrm{Hom}(\pi_1(X), \pi)$.

3. For a space X let $\mathrm{Aut}(X)$ denote the group of homotopy classes of homotopy equivalences $X \to X$. Show that for a CW complex $K(\pi,1)$, $\mathrm{Aut}\big(K(\pi,1)\big)$ is isomorphic to the group of outer automorphisms of π, that is, automorphisms modulo inner automorphisms.

4. With the notation of the preceding problem, show that $\mathrm{Aut}(\bigvee_n S^k) \approx \mathrm{GL}_n(\mathbb{Z})$ for $k > 1$, where $\bigvee_n S^k$ denotes the wedge sum of n copies of S^k and $\mathrm{GL}_n(\mathbb{Z})$ is the group of $n \times n$ matrices with entries in $\mathbb{Z}$ having an inverse matrix of the same form. [$\mathrm{GL}_n(\mathbb{Z})$ is the automorphism group of $\mathbb{Z}^n \approx \pi_k(\bigvee_n S^k) \approx H_k(\bigvee_n S^k)$.]

5. This problem involves the spaces constructed in the latter part of this section.
(a) Compute the homology groups of the complex Z in the case $n = 3$, when Z is 2-dimensional.
(b) Letting $\widetilde{X}_n$ denote the n-dimensional complex $\widetilde{X}$, show that $\widetilde{X}_n$ can be obtained inductively from $\widetilde{X}_{n-1}$ as the union of two copies of the mapping torus of the generating deck transformation $\widetilde{X}_{n-1} \to \widetilde{X}_{n-1}$, with copies of $\widetilde{X}_{n-1}$ in these two mapping tori identified. Thus there is a fiber bundle $\widetilde{X}_n \to S^1 \vee S^1$ with fiber $\widetilde{X}_{n-1}$.
(c) Use part (b) to find a presentation for $\pi_1(\widetilde{X}_n)$, and show this presentation reduces to a finite presentation if $n > 2$ and a presentation with a finite number of generators if $n = 2$. In the latter case, deduce that $\pi_1(\widetilde{X}_2)$ has no finite presentation from the fact that $H_2(\widetilde{X}_2)$ is not finitely generated.

4.B The Hopf Invariant

In §2.2 we used homology to distinguish different homotopy classes of maps $S^n \to S^n$ via the notion of degree. We will show here that cup product can be used to do something similar for maps $S^{2n-1} \to S^n$. Originally this was done by Hopf using more geometric constructions, before the invention of cohomology and cup products.

In general, given a map $f : S^m \to S^n$ with $m \geq n$, we can form a CW complex C_f by attaching a cell e^{m+1} to S^n via f. The homotopy type of C_f depends only on the homotopy class of f, by Proposition 0.18. Thus for maps $f, g : S^m \to S^n$, any invariant of homotopy type that distinguishes C_f from C_g will show that f is not homotopic to g. For example, if $m = n$ and f has degree d, then from the cellular chain complex of C_f we see that $H_n(C_f) \approx \mathbb{Z}_{|d|}$, so the homology of C_f detects the degree of f, up to sign. When $m > n$, however, the homology of C_f consists of $\mathbb{Z}$'s in dimensions 0, n, and $m+1$, independent of f. The same is true of cohomology groups, but cup products have a chance of being nontrivial in $H^*(C_f)$ when $m = 2n-1$. In this case, if we choose generators $\alpha \in H^n(C_f)$ and $\beta \in H^{2n}(C_f)$, then the multiplicative structure of $H^*(C_f)$ is determined by a relation $\alpha^2 = H(f)\beta$ for an integer $H(f)$ called the **Hopf invariant** of f. The sign of $H(f)$ depends on the choice of the generator β, but this can be specified by requiring β to correspond to a fixed generator of $H^{2n}(D^{2n}, \partial D^{2n})$ under the map $H^{2n}(C_f) \approx H^{2n}(C_f, S^n) \to H^{2n}(D^{2n}, \partial D^{2n})$ induced by the characteristic map of the cell e^{2n}, which is determined by f. We can then change the sign of $H(f)$ by composing f with a reflection of S^{2n-1}, of degree -1. If $f \simeq g$, then under the homotopy equivalence $C_f \simeq C_g$ the chosen generators β for $H^{2n}(C_f)$ and $H^{2n}(C_g)$ correspond, so $H(f)$ depends only on the homotopy class of f.

If f is a constant map then $C_f = S^n \vee S^{2n}$ and $H(f) = 0$ since C_f retracts onto S^n. Also, $H(f)$ is always zero for odd n since in this case $\alpha^2 = -\alpha^2$ by the commutativity property of cup product, hence $\alpha^2 = 0$.

Three basic examples of maps with nonzero Hopf invariant are the maps defining the three Hopf bundles in Examples 4.45, 4.46, and 4.47. The first of these Hopf maps is the attaching map $f : S^3 \to S^2$ for the 4-cell of CP^2. This has $H(f) = 1$ since $H^*(\mathbb{C}\mathrm{P}^2; \mathbb{Z}) \approx \mathbb{Z}[\alpha]/(\alpha^3)$ by Theorem 3.12. Similarly, $\mathbb{H}\mathrm{P}^2$ gives rise to a map $S^7 \to S^4$ of Hopf invariant 1. In the case of the octonionic projective plane $\mathbb{O}\mathrm{P}^2$, which is built from the map $S^{15} \to S^8$ defined in Example 4.47, we can deduce that $H^*(\mathbb{O}\mathrm{P}^2; \mathbb{Z}) \approx \mathbb{Z}[\alpha]/(\alpha^3)$ either from Poincaré duality as in Example 3.40 or from Exercise 5 for §4.D.

It is a fundamental theorem of [Adams 1960] that a map $f : S^{2n-1} \to S^n$ of Hopf invariant 1 exists only when $n = 2, 4, 8$. This has a number of very interesting consequences, for example: